AI-Powered Pricing Assistant

Project Overview

We require a locally hosted AI-powered software solution that can simply speak about the data that we feed into the machine . The purpose of the software is to enable authorized users to retrieve pricing, and business-related information using natural language voice or text commands while also generating business insights and performing approved calculations.

The solution should run primarily on our local infrastructure to ensure data privacy and security.

Objectives

1. Extract and synchronize inventory and pricing data from BUSY / OUR EXCEL SHEETS
2. Allow users to interact with the system using voice and text.
3. Provide instant responses regarding product prices, stock availability, margins, and related business information.
4. Generate business insights and trends from historical data.
5. Support calculation-based commands such as discounts, margin calculations, price revisions, etc.
6. Restrict access to authorized users only.
7. Allow continuous data updates as product prices and inventory change.

Core Features

1. Excel , Pdf data extraction

The system must:

- Read data from our excel sheets to extract buying and selling values

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2. Voice-Based AI Assistant (HINDI + ENGLISH)

Users should be able to interact using voice commands.

Examples:

"What is the selling price of Tenax Tile Cleaner?"

"How much stock is available for Product X?"

"Who is the supplier for Product Y?"

"What was our last purchase rate for Product Z?"

The assistant should respond via:

- Voice
- On-screen text

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3. Text-Based Assistant

Users should also be able to type queries.

Examples:

- Show all products below reorder level.
- Which products have not sold in the last 60 days?
- Show highest margin products.
- Compare current month sales with last month.

Security Requirements

- Login authentication
- Password protection
- Audit logs for all modifications
- Optional biometric authentication in future versions

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8. Local Deployment

Primary requirement:

- Entire solution should run locally.
- Data should remain within company infrastructure.

Preferred deployment:

- Windows Desktop Application
  OR
- Local Web Application hosted on company server.
- **IT MUST BE COMPATIBLE WITH ANDROID THE REASON BEING THE MAJORITY IT WILL BE USED ON ANDROID MOBILE PHONES .**

No mandatory cloud dependency.

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9. Data Update Management

The system must be able to :

- detect updated prices.
- Track changes over time.

Example:

If product purchase price changes from ₹200 to ₹220, the system should maintain history and use updated values for future calculations.

Technical Expectations

Preferred Technology Stack (Open to Suggestions)

Backend:

- Python
- FastAPI

Database:

- PostgreSQL or SQLite

AI Layer:

- Local LLM (Qwen, Llama, Mistral, etc.)

- Retrieval-Augmented Generation (RAG)

Voice:

- Speech-to-Text
- Text-to-Speech

Frontend:

- React
- Electron Desktop App OR Web Dashboard

Deliverables

Developer/Agency should provide:

1. System Architecture Document
2. UI/UX Design
3. AI Assistant Module
4. Voice Interface
5. Dashboard & Reporting
6. User Management System
7. Deployment Setup In our Phones and Laptop by developer
8. Documentation
9. Source Code Ownership Transfer

Success Criteria

The solution should allow an authorized user to ask inventory, pricing, stock, sales, and business-related questions in natural language and receive accurate responses in under 5 seconds while maintaining complete data privacy and local deployment.

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Additional Functional Requirements

1. Role-Based Authentication and Access Control

The system shall implement a secure role-based authentication mechanism with separate access levels for different types of users. Access to sensitive business information shall be controlled based on the authenticated user's role.

Owner / Administrator Access

The Owner or Administrator shall have unrestricted access to all business data and system functionalities, including but not limited to:

- Buying (landing) prices of all products.
- Selling prices of all products.
- Profit margins and pricing analysis.
- Supplier information and purchase history.
- Stock availability and inventory reports.
- Business analytics and dashboards.
- Product management (Add, Edit, Delete).
- User management and permission settings.
- AI configuration and system settings.

Staff Access

Staff members shall have restricted access to operational information required for their daily tasks. Staff users shall be permitted to:

- View only the selling prices of products.
- Search products using text or voice commands.
- Check stock availability (subject to business policy).
- Access customer-facing information only.

Staff users shall **not** have access to:

- Buying (landing) prices.
- Profit margins.
- Supplier information.
- Financial reports.
- Administrative settings.
- User management features.

The AI assistant shall strictly enforce these permission levels. If a user requests information outside their authorized access level, the system shall deny the request and respond with an appropriate permission-related message without exposing any restricted data.

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2. Secure Remote Accessibility

The application shall not be limited to operation within a local Wi-Fi network. It shall support secure remote access from any authorized location over the internet while maintaining the confidentiality and integrity of business data.

The system shall enable users to securely access the application from:

- Mobile phones.
- Tablets.
- Laptops.
- Desktop computers.

Authorized users should be able to access the system from:

- Retail shops.
- Warehouses.
- Construction sites.
- Market locations.
- Remote offices.
- Any other location with an internet connection.

The office computer or central server shall act as the primary data source, ensuring that all connected devices retrieve real-time information from a single centralized database.

The system shall provide:

- Secure encrypted communication between client devices and the server.
- User authentication before granting access.
- Real-time synchronization of inventory, pricing, and business data.
- Simultaneous support for multiple users without data inconsistency.
- Reliable performance over both local and remote internet connections.

This architecture shall allow business operations to continue seamlessly regardless of the user's physical location.

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3. Mobile-Friendly Progressive Web Application (PWA)

The application shall be developed as a Progressive Web Application (PWA) to provide a mobile app-like experience without requiring installation through traditional app stores.

The application shall support installation directly from a supported web browser using the "Add to Home Screen" feature.

Once installed, the application shall:

- Display its own application icon on the device's home screen.
- Launch in a full-screen interface similar to a native mobile application.
- Provide a fast and responsive user experience.
- Automatically adapt to different screen sizes and device orientations.
- Support touch-friendly navigation and voice interaction.
- Maintain consistent functionality across Android, iOS, Windows, and macOS devices.

The application should also support:

- Automatic updates without requiring manual reinstallation.
- Background caching of essential application assets to improve loading performance.
- Future integration with push notifications and additional mobile-specific capabilities.

This approach shall allow users to access the AI assistant quickly from their mobile home screen while eliminating the need to publish or maintain separate Android and iOS applications.